utility still applied when the probabilities were subjective and derived from agents' actions, rather than being taken as given. Thus, the probabilities in equation (2.2) can be objective or subjective.[9]

To make choices, the asset owner *maximizes expected utility*. The problem is formally stated as:

$$\max_{\theta} E[U(W)], \tag{2.3}$$

where θ are choice (or control) variables. Some examples of choice variables include asset allocation decisions (portfolio weights), spending/savings plans, and production plans for a firm, and so on. The maximization problem in equation (2.3) is often solved subject to constraints. For an asset owner, these constraints often involve constraints on the *investment universe*, *position constraints* like *leverage constraints*, no *short-sale constraints*, or the inability to hold certain sectors or asset classes, and, when the asset allocation problem is being done over time, there may be constraints that depend on an investor's past asset positions (like *turnover constraints*).

A very special class of the expected utility framework is the set of *rational* expected utility models. Traditionally, these were derived to satisfy certain axioms (like the independence axiom, which states roughly that if I prefer x to y, then I also prefer the possibility of getting x to the possibility of getting y when x and y are mixed with the same lottery; violation of this maxim led to the loss aversion utility that I describe in section 4 of this chapter). Not surprisingly many of these axioms are violated in reality.[10] Rejecting the whole class of expected utility as a decision-making tool solely on this restrictive set of rational expected utility models is misguided; expected utility remains a useful tool to guide asset owners' decisions—even when these investors have behavioral tendencies, as most investors do in the real world. In fact, the broad class of expected utility nests behavioral models, as I describe below.

2.5. WHAT CHOICE THEORY IS NOT ABOUT

How agents make decisions, and giving them advice to make good decisions, is not about

1. Wealth

    Greater wealth gives asset owners more choices. But wealth per se is not the objective. Figure 2.1 illustrates that how we get to a given level of wealth—through slow and steady increases or by extreme ups and downs—can matter

[9] See Schoemaker (1982) for a summary on the expected utility model and its variants.

[10] An excellent book that offers insight into investors' behavioral tendencies is Kahneman's (2011) *Thinking Fast and Slow*.